

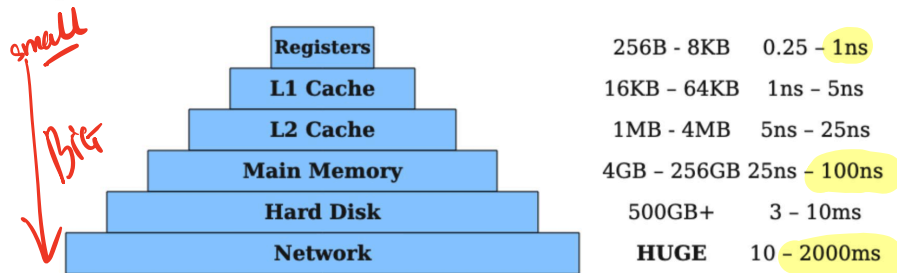
CSE 231: Register Allocation

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Memory Hierarchy ...

Not all *memory* is created equal!

faster, smaller



slower, bigger

(via Max New)

... Stack is an Expensive Place to Store Values

Source

```
(let ((a0 92)
      (a1 (add1 a0))
      (a2 (add1 a1))
      (a3 (add1 a2))
      (a4 (add1 a3))
      (a5 (add1 a4)))
  a5)
```

"bad" Asm

```
mov rax, 184
mov [rbp - 8*2], rax
mov rax, [rbp - 8*2]
add rax, 2
mov [rbp - 8*3], rax
mov rax, [rbp - 8*3]
add rax, 2
mov [rbp - 8*4], rax
mov rax, [rbp - 8*4]
add rax, 2
mov [rbp - 8*5], rax
mov rax, [rbp - 8*5]
add rax, 2
mov [rbp - 8*6], rax
mov rax, [rbp - 8*6]
add rax, 2
mov [rbp - 8*7], rax
mov rax, [rbp - 8*7]
```

"good" Asm-Opt

```
mov rbx, 184
add rbx, 2
add rbx, 2
add rbx, 2
add rbx, 2
add rbx, 2
add rbx, 2
mov rax, rbx
```

Midterm IC
TOMOROW 3P

EGG →
FDL → scores later

Much faster to access registers than main memory!

```
$ time ./test/reg_slow.fun.run 1000000000
97
Executed in 1.84 secs ...

$ time ./test/reg_opt.fun.run 1000000000
97
Executed in 763.37 millis ...
```

Optimization: Register Allocation

Lets use **registers** instead of defaulting to **stack** storage.

Why is this tricky?

Few - registers

"over writing"

R1 ... R7

QUIZ: Register Optimized Code

Program

```
(let ((a1 (+ 10 10))
      (a2 (* 2 a1))
      (a3 (* 3 a2)))
  (* 10 a3))
```

Optimized Asm

```
mov rax, <10>
add rax, <10>
mul rax, <2>
mul rax, <3>
mul rax, <10>
```

Alloc

$a_1 \rightarrow RAX$

$a_2 \rightarrow RAX$

$a_3 \rightarrow RAX$

```
(let ((n (* 5 5))
      (m (* 6 6))
      (x (+ n 1))
      (y (+ m 1)))
  (+ x y))
```

```
mov rax, 5
mul rax, 5
mov rbx, 6
mul rbx, 6
add rax, 1
add rbx, 1
add rax, rbx
```

$n, x \rightarrow RAX$

$m, y \rightarrow RBX$

$n, x \rightarrow RAX$

$m, y \rightarrow \text{Stack}(1)$

```
(defn (f a)
  (let ((x (* a 2))
        (y (+ x 7)))
    y))
```

```
; a --> [rbp + 16]
mov rax, [rbp+16]
mul rax, 2
add rax, 7
```

```
(defn (f a)
  (let ((x (* a 2))
        (y (+ x 7)))
    (g x y)))
```

```
; a --> [rbp + 16]
```

Optimization Relies on Variables?

What if the programmer *instead* wrote code like

Example 1

```
(* 10 (* 3 (* 2 (+ 10 10))))
```

Example 2

```
(+ (+ (* 5 5) 1) (+ (* 6 6) 1))
```

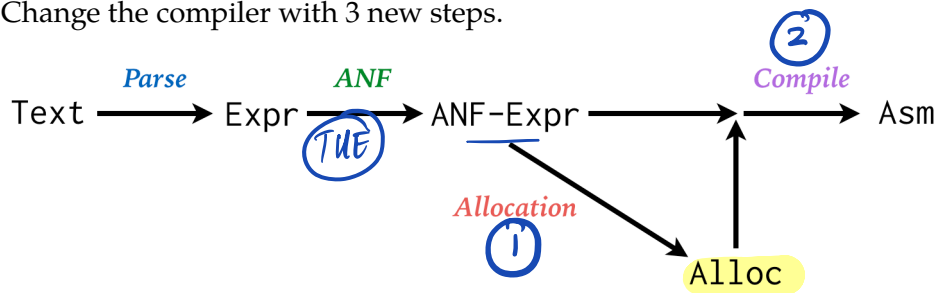
Example 3

```
(defn (f a)
  (+ (* a 2) 7))
```

Yikes, how to optimize without *names*?

Register Optimization Pipeline

Change the compiler with 3 new steps.



1. **Transform** to ANF *intermediate representation*,
2. **Allocation** of variables to *registers*,
3. **Compile** using *allocation*.

Administrative Normal Form (ANF) Transform

- **Immediate:** *constants or variable lookups* whose values can be loaded with a single machine instruction
- **ANF:** every call or prim-op's arguments are *immediate*.

Step 1 transforms $(+ \overset{x}{(+ \overset{n}{(* 5 5)} 1)} \overset{y}{(+ \overset{m}{(* 6 6)} 1)})$ to ANF

```
(let ((n (* 5 5))
      (m (* 6 6))
      (x (+ n 1))
      (y (+ m 1)))
  (+ x y))
```

A blue arrow points from the expression $(+ (+ (* 5 5) 1) (+ (* 6 6) 1))$ in the text above to the 'let' block in the code below.

Lets look at these steps *backwards* starting with Step 3 (compilation), then Step 2 (allocation), and finally, ANF transform lecture.

Step 3: Allocation (Assigning Vars to Registers)

Registers: Fast Storage for Variables

Some fixed set of registers that we can use to store variables.

```
enum Reg {
    RAX, RBX, RCX, RDX,
    R8, R9, R10, R11, ...
}
```

Locations: Where to Store a Variable

A Loc is a *location* where a variable can be stored,

- a *register* or
- a *stack offset*

```
pub enum Loc {
    /// Register
    Reg(Reg),
    /// Stack local: [rbp - 8 * offset]
    Stack(i32),
}
```

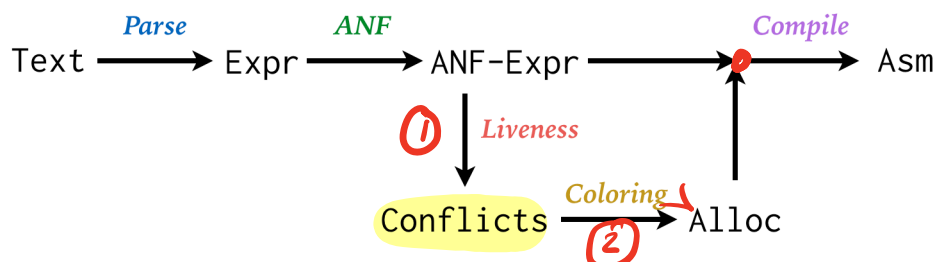
Allocation: Mapping Variables to Locations

An Alloc is a mapping from variable names to Loc:

```
pub struct Alloc(HashMap<String, Loc>);
```

Computing an Allocation

How to compute Alloc for a given function body (Expr)?



Step 1: Build *conflict graph* over function variables

Step 2: Color variables so *adjacent vars have different colors*.

1. Conflict Graph

Undirected graph where:

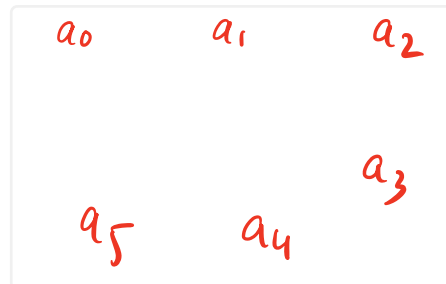
- Vertices are *variables*
- Edges between x, y if both *must be available at same time*

QUIZ: Fill in the Conflict Graphs

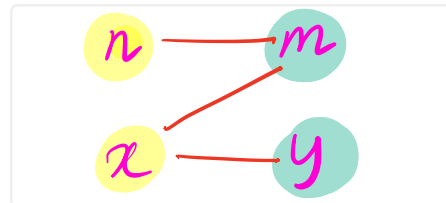
Program

```
(let ((a0 92))
  (a1 (+ a0 1))
  (a2 (+ a1 1))
  (a3 (+ a2 1))
  (a4 (+ a3 1))
  (a5 (+ a4 1)))
a5)
```

Conflict Graph



```
(let ((n (* 5 5))
      (m (* 6 6))
      (x (+ n 1))
      (y (+ m 1)))
  (+ x y))
```



2. Coloring

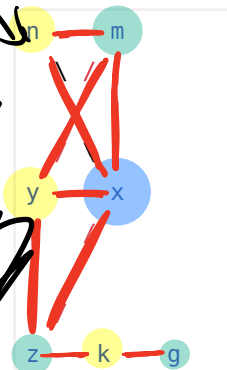
- Label each *vertex* with a **color** (register)
- Ensuring **adjacent** vertices have **different** colors.

Program

```
(let ((n (* 5 5))
      (m (* 6 6))
      (x (+ n 1))
      (y (+ m 1))
      (z (+ x 1))
      (g (+ k 5)))
  (+ x y))
```

naughty code generator

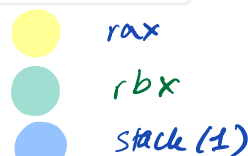
Conflict Graph



Coloring

VAR	REG
n	----
m	----
x	----
y	----
z	----
k	----
g	----

What if we require n colors, but only have $r < n$ registers? *Spill* the remaining $n - r$ variables onto the stack!



Conflict Graph from Live Variables

We stared at the code to find "conflicts". How to *automate*?

Vars $\{x_1 \dots x_n\}$ are **live** for e if we *need* the values of $x_1 \dots x_n$ to evaluate e .

1. Identify **live variables** at each point,
2. Conflicts between **pairs of concurrently live variables**.

QUIZ: Live Variables

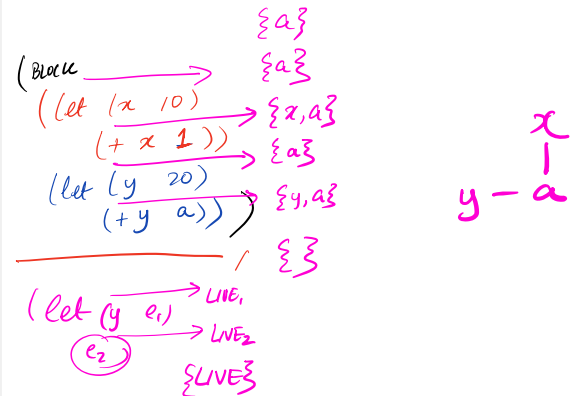
Lets identify the live/conflicts in the quiz above.

```

;      LIVE Variables

;
(let (a0 92)
; → {a0}
(let (a1 (+ a0 1))
; → {a1}
(let (a2 (+ a1 1))
; → {a2}
(let (a3 (+ a2 1))
; → {a3}
(let (a4 (+ a3 1))
;
(let (a5 (+ a4 1))
;
a5))))))

```

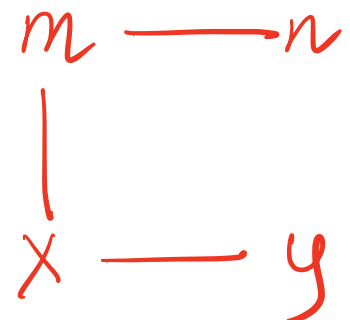


```

;      LIVE Variables

; → {}
(let (n (* 5 5))
; → n
(let (m (* 6 6))
; → m, n
(let (x (+ n 1))
; → m, n
(let (y (+ m 1))
; → x, y
(+ x y))))))
; → {}

```



Similar to **free variables** for closure (but not quite same...)

Implementing Live in your compiler

Your compiler will implement a function

```

fn live(
  graph: &mut ConflictGraph, /// graph of edges
  e: &Expr,                  /// expression to analyze
  binds: &HashSet<String>,    /// let-binds outside e
  params: &HashSet<String>,    /// funparams (on stack)
  out: &HashSet<String>,      /// vars "LIVE" after `e`
) -> HashSet<String>          /// vars "LIVE" *before*

```

graph is mut: Add *conflict edges* while traversing e.

(defn (foo x₁ x₂ x₃)
body)

Step 2: Compilation

Use Allocated Registers (not only *rax*)

Suppose that

- *imm1* and *imm2* immediate values,
- Alloc maps *imm1* to *<imm1>* and *imm2* to *<imm2>*,
- we want to compile *(+ imm1 imm2)* to *<dst>*.

How to compile ANF code using the Alloc mapping?

ANF Code

```
(+ imm1 imm2)
```

Compiled Asm

```
mov <dst>, <imm1>
add <dst>, <imm2>
```

What if *dst* is on the stack?
What if *dst* is a register?

```
(let ((a0 92))
  (a1 (+ a0 1))
  (a2 (+ a1 1))
  (a3 (+ a2 1))
  (a4 (+ a3 1))
  (a5 (+ a4 1)))
a5)
```

```
; ALLOC a0...a5 --> rax
mov rax, 92
add rax, 1
...
add rax, 1
```

Saving Registers

Callee functions can overwrite caller registers!

1. Callee saves Each function saves & restores all of the registers it uses; callers do not have to store anything.

2. Caller saves Each caller saves & restores all of the registers it uses; callees do not have to store anything.

3. Hybrid Combine the two options; some registers are caller-saved others are callee-saved

Starter code does "callee save"; beware calls into runtime.

